

HYESAN, Korea, Democratic People's Republic of

WMO: 470160

Lat: 41.3998N

Lon: 128.1796E

Elev: 714

StdP: 93.04

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.8	-24.9	-29.8	0.3	-26.0	-27.9	0.3	-24.1	7.3	-15.4	5.9	-13.6	0.4	50	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	30.2	20.6	28.7	19.9	27.2	19.4	22.7	27.9	21.9	26.5	21.0	25.2	1.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	17.3	25.4	20.4	16.4	24.5	19.6	15.6	23.6	71.1	27.9	67.7	26.5	64.6	25.3	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	4.5	3.5	DB	-30.3	33.0	1.6	1.6	-31.5	34.1	-32.4	35.0	-33.2	35.9	-34.4	37.1	
			WB	-30.3	24.5	1.6	1.6	-31.5	25.7	-32.4	26.7	-33.2	27.6	-34.4	28.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-15.6	-11.2	-2.7	6.2	13.1	18.1	21.8	20.9	14.5	6.7	-12.8		
		DBStd	13.57	4.35	5.05	5.05	4.44	3.52	2.69	2.36	2.84	3.33	4.08	5.56	4.81	
		HDD10.0	3163	794	594	393	129	13	0	0	4	116	412	707		
		HDD18.3	5217	1053	827	652	364	166	36	3	10	119	360	662	966	
		CDD10.0	1222	0	0	0	14	108	243	365	337	141	14	0	0	
		CDD18.3	234	0	0	0	0	2	29	109	89	6	0	0	0	
		CDH23.3	1776	0	0	0	10	104	315	746	549	50	2	0	0	
		CDH26.7	443	0	0	0	0	19	68	220	132	4	0	0	0	
(14)	Wind	WSAvg	0.7	0.6	0.8	1.0	1.0	1.0	0.7	0.5	0.5	0.5	0.6	0.5	0.5	
(15)	Precipitation	PrecAvg	648	7	11	15	33	69	100	164	118	61	29	26	14	
(16)		PrecMax	852	21	39	31	68	153	176	328	216	143	55	70	48	
(17)		PrecMin	426	1	2	3	10	25	26	47	28	7	2	4	2	
(18)		PrecStd	105	5	8	8	15	28	40	63	47	37	16	15	11	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	-1.0	4.8	14.7	24.8	28.9	30.5	33.4	32.0	27.1	21.9	11.8	0.9		
(20)		0.4% MCWB	-2.6	1.4	7.8	13.6	15.9	18.4	22.2	22.0	18.4	12.8	6.3	-1.4		
(21)		2% DB	-4.2	2.0	10.8	20.9	26.1	28.6	30.7	29.6	24.8	18.9	8.6	-1.4		
(22)		2% MCWB	-5.9	-0.8	4.7	10.6	14.6	17.3	21.4	21.5	16.9	11.0	4.6	-3.1		
(23)		5% DB	-6.4	-0.1	8.1	18.1	23.9	26.9	29.0	28.0	23.0	16.5	6.1	-3.4		
(24)		5% MCWB	-7.8	-2.6	3.1	9.2	13.4	17.3	21.0	20.9	15.9	9.6	3.1	-5.0		
(25)		10% DB	-8.4	-2.1	5.5	15.1	21.6	25.0	27.4	26.5	21.3	14.2	3.9	-5.3		
(26)		10% MCWB	-9.6	-4.1	1.6	7.4	12.5	16.7	20.5	20.1	15.1	8.5	1.6	-6.6		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	-2.7	1.7	8.2	14.5	18.1	20.7	24.1	23.8	20.4	13.9	7.4	-0.7		
(28)		0.4% MCDB	-1.1	3.8	13.3	21.7	24.4	26.6	30.0	29.6	24.0	20.2	10.2	0.5		
(29)		2% WB	-5.8	-0.8	5.3	11.6	16.0	19.3	23.0	22.7	18.7	12.0	5.1	-3.0		
(30)		2% MCDB	-4.3	1.6	9.9	19.0	22.7	25.2	28.5	27.6	22.6	17.4	8.1	-1.5		
(31)		5% WB	-7.8	-2.5	3.4	9.7	14.6	18.3	22.1	21.9	17.2	10.5	3.4	-4.9		
(32)		5% MCDB	-6.5	-0.4	7.3	16.6	22.1	24.1	27.1	26.3	21.1	15.3	5.7	-3.5		
(33)		10% WB	-9.6	-4.2	1.7	8.1	13.4	17.5	21.3	21.2	16.1	9.1	1.6	-6.6		
(34)		10% MCDB	-8.4	-2.1	5.2	14.4	20.1	23.1	25.8	25.1	19.7	13.2	3.7	-5.2		
(35)	Mean Daily Temperature Range	MDBR	12.4	13.4	12.0	13.1	13.7	12.3	10.2	10.2	12.6	12.9	10.6	11.2		
(36)		5% DB MCDBR	12.7	14.0	15.2	18.9	19.0	16.4	13.7	12.4	14.6	16.6	13.4	11.4		
(37)		5% WB MCWBR	11.2	11.3	10.0	10.2	9.1	7.1	5.9	5.6	7.5	9.5	9.4	9.8		
(38)		5% WB MCDBR	12.6	13.1	13.8	17.1	16.6	13.2	11.1	10.3	11.8	14.8	12.2	11.1		
(39)	Clear-Sky Solar Irradiance	taub	0.328	0.389	0.467	0.541	0.576	0.594	0.531	0.481	0.427	0.428	0.374	0.333		
(40)		taud	2.079	1.932	1.802	1.726	1.710	1.738	1.972	2.101	2.168	2.037	2.072	2.085		
(41)		Ebn at Noon	788	784	763	740	726	714	758	785	802	737	722	741		
(42)		Edh at Noon	110	150	193	223	233	227	178	152	132	134	111	101		
(44)	All-Sky Solar Radiation	RadAvg	2.48	3.46	4.48	4.87	5.24	5.02	4.67	4.58	4.26	3.34	2.27	2.04		
(45)		RadStd	0.14	0.16	0.26	0.30	0.47	0.48	0.49	0.40	0.40	0.29	0.23	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page